# 0.5 Project Scope Sheet · Group 1 · Fernão de Oliveira

11 November 2026 · T. Santos, L. Batista

**Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group**

| Field | Value |
|---|---|
| Group | Group 1 · Fernão de Oliveira |
| Members (student number) | 106238 · 107304 · 106645 |
| Baseline | B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week |
| Specialization 1 | SP01 · Longitudinal Strength Calculation |
| Specialization 2 | SP02 · Propulsion Matching |
| Deadline | Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com |
| ZIP name | Group1_Project.zip |
| Presentation | Dec 18, 2026, 15 minutes per group |

**Planning (advised).** A work plan is not a deliverable, but plan internally: a rough approach and a split of the tasks across the group, so that the work, the presentation and the knowledge of the whole project are shared evenly by every member. You are welcome to share the plan and discuss it with the instructor.

## Deliverable Groups D1–D11

| # | Deliverable | Your project must deliver (group, outside class) |
|---|---|---|
| D1 | Design Basis | 130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked |
| D2 | Reference Ship Database | At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds |
| D3 | Ship Model | Volume-carrier sequence · inputs, outputs and calculation sequence · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · electric drive, two shafts · batteries as deadweight · crane in the lightship · closed weight balance |
| D4 | Voyage and Financial Model | Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input |
| D5 | Validation Note | Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed |
| D6 | Optimization Report | Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements |
| D7 | Compliance Sheet | EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules (BV) · stability while the crane lifts · freeboard · draught at every port |
| D8 | General Arrangement and Hull Form | Deck container layout and crane reach · battery slots aft · hull form of your choice, justified (a PSV-type hull is only a suggestion) · A3 sketch (profile and midship section) · hydrostatics check |
| D9 | Specialization Documents (2) | One calculation note per specialization, with its required drawings (see Your specializations) |
| D10 | Datasheet and Presentation | One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · presentation of 15 minutes |

| # | Deliverable | Your project must deliver (group, outside class) |
|---|---|---|
| D11 | Code Flowchart (mandatory) | A flowchart of your code that shows how the information flows: inputs, each calculation step, the iterations and the outputs · symbols of ISO 5807, the standard for data, program and system flowcharts |

Exercises 7.1 and 7.2 (digital transition and AI) come after the deadline: your group may show them in the presentation, but they are not part of the ZIP.

## Your studies and drawings

Your group's list from the Statement of Work: 33 items, 5 of them drawings. Tick each one off as it is done.

| ID | Title | Deliverable | Status |
|---|---|---|---|
| STU-D1.1 | Owner's requirements and port limits | D1 | |
| STU-D1.2 | Environmental targets | D1 | |
| STU-D2.1 | Reference ship database with sources | D2 | |
| STU-D2.2 | Regressions and bounds | D2 | |
| STU-D3.1 | Ship model | D3 | |
| STU-D3.2 | Closed weight balance | D3 | |
| STU-D4.1 | Voyage model | D4 | |
| STU-D4.2 | Financial model | D4 | |
| STU-D5.1 | Validation against reference ships | D5 | |
| STU-D6.1 | Optimization | D6 | |
| STU-D7.1 | EEDI and CII | D7 | |
| STU-D7.2 | Freeboard and intact stability | D7 | |
| STU-D7.3 | Draught at every port | D7 | |
| STU-D8.1 | Hydrostatics check | D8 | |
| DWG-D8.1 | Profile (A3) | D8 | |
| DWG-D8.2 | Midship section (A3) | D8 | |
| STU-D10.1 | Datasheet (template 0.4) | D10 | |
| STU-D10.2 | Presentation, 15 minutes | D10 | |
| STU-D11.1 | Code flowchart (ISO 5807) | D11 | |
| STU-B.1 | Capacity count on own arrangement | D8 | |
| DWG-B.1 | Container and battery layout with crane reach | D8 | |
| STU-SP01.1 | Lightship and deadweight distribution | D9 · SP01 | |
| STU-SP01.2 | Four critical loading conditions | D9 · SP01 | |
| STU-SP01.3 | Still-water and wave bending moments and shear forces | D9 · SP01 | |
| STU-SP01.4 | Allowable stresses, bending and shear | D9 · SP01 | |
| STU-SP01.5 | Preliminary scantlings of the midship cross-section | D9 · SP01 | |
| STU-SP01.6 | Section modulus, moment of inertia and neutral axis | D9 · SP01 | |
| STU-SP01.7 | Allowable still-water bending moment and shear force vs loading conditions | D9 · SP01 | |

| ID | Title | Deliverable | Status |
|---|---|---|---|
| DWG-SP01.1 | Midship cross-section | D9 · SP01 | |
| STU-SP02.1 | Optimum B-series propeller | D9 · SP02 | |
| STU-SP02.2 | Engine matching | D9 · SP02 | |
| STU-SP02.3 | PTO/PTI or electric operating modes | D9 · SP02 | |
| DWG-SP02.1 | Stern arrangement | D9 · SP02 | |

## Your specializations

One calculation note per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

### SP01 · Longitudinal Strength Calculation

Show that the hull girder carries the still-water and wave bending moments.

- STU-SP01.1 Lightship weight and deadweight distribution along the length.
- STU-SP01.2 Four critical loading conditions (subject to justification and/or definition with the instructor).
- STU-SP01.3 Still-water and wave bending moments and shear forces for each loading condition.
- STU-SP01.4 Allowable stresses, in bending and in shear.
- STU-SP01.5 Preliminary scantlings of the midship cross-section.
- STU-SP01.6 Section modulus, moment of inertia and neutral axis of the midship section.
- STU-SP01.7 Allowable vertical still-water bending moment and shear force: the allowable envelopes along the length, with each loading condition plotted against them.
- DWG-SP01.1 Midship cross-section.

### SP02 · Propulsion Matching

Choose the propeller and match it to the motors in every operating mode.

- STU-SP02.1 Optimum Wageningen B-series propeller for the service speed.
- STU-SP02.2 Engine matching: operating points on the motor layout diagram, with the sea margin.
- STU-SP02.3 The two-shaft electric modes (Baseline B).
- DWG-SP02.1 Stern arrangement: propeller, shaft line and rudder.

The full text of every item is in the Statement of Work.